CONSTRUCTOR TECHNOLOGY · EEG MIND READING

Telepathy

Yetkin Zambelli | Mike Mengjie Huang

Decoding visual perception
from EEG signals

Hackathon Pitch · October 2026



ML PIPELINE

Preprocessing → Features → Classification

01	**Preprocessing**	Bandpass 0.5–75 Hz + 50 Hz notch
02	**EEG Features**	LaBraM, modified LaBraM (3s), LUNA, CBraMod
03	**Video Features**	OpenCLIP, DINOv2, AlexNet
04	**Classification**	Per-subject RidgeCV: map video features → EEG response, rank by Pearson

RESULTS

Test retrieval (102 candidates)

Chance: Top-1 = 0.98% · Top-5 = 4.9%

Subject	Top-1	Top-5	Median Rank
01	5.9%	24.5%	15.0
02	3.9%	24.5%	19.5
03	4.9%	17.7%	20.5
04	4.9%	14.7%	22.5

FEATURE ABLATION

Subject 1

Features	Test Top-1	Test Top-5	Median Rank
AlexNet features.3	3.9%	11.8%	25.5
DINO blocks.6	4.9%	30.4%	13.0
CLIP resblocks.6	3.9%	16.7%	17.0
CLIP resblocks.7	4.9%	21.6%	14.0
Ensemble (submitted)	5.9%	26.5%	14.0

Why Ridge on raw EEG won

CCA	100% train top-1, but test at chance (median rank ~50)
CLIP-style contrastive	Heavy overfitting; memorizes training pairs
MLP regression	Underfits; train already at chance
Ridge (submitted)	Only method clearly above chance on test

KEY TAKEAWAYS

What we learned

- Strong regularization ($\alpha \approx 10^4$–10^5) was critical for generalization
- Ensemble of video features beat any single source on test top-1
- Learned EEG embeddings (LaBraM, LUNA, CBraMod) overfit more easily
- DINO features gave the strongest single-model performance

Telepathy

Decoding what the mind sees

Thank you!

